UTANG TRACKER:
A Customer Credit and Payment Recording System For Sari-Sari Store Owner

A Project Presented to the Faculty of
the College of Information Technology
Cagayan de Oro College – Phinma Education

In Partial Fulfillment of the Requirement for the Course
ITE 300 – Object Oriented Programming

Submitted by:

EBARAT, REINZ B.

MAAMBONG, RESCLE

MANGA, DEBORAH D.

OGUIMAS, AERON JHON

PAJE, JERICHO

COC-FAB-BSIT2-07

Submitted to:
Faith Evangeline M. Dollera, MIT

October 2026

Table of Contents

Part 1 - Introduction

Part 2 - User’s Functionality

Part 3 - System Process Flow Diagram

Part 4 - Code Snippet (Arranged by Users and System Flow)

Part 5 - Graphical User Interface (Screenshots)

Part 6 - Moments with Team

Part 1

Introduction

Background of the Study

Sari-sari stores in Igpit, Opol, Misamis Oriental still record utang in notebooks. A notebook is easy to lose, hard to search, and does not separate the person who borrowed from the person who recorded the sale. Utang Tracker is a Java desktop system built for ITE 300 to replace that notebook with one shared record for the owner, the staff, and the customers.

The system has three users. Customers register with a separate first name and last name, a contact number, a username, a password, and a 4-digit recovery PIN, each confirmed by typing it twice. A new account stays pending until the owner checks the customer's valid ID and approves it. Once approved, the customer can view only their own balance and history. The staff searches for a customer by ID or name, adds utang, and records payments. The owner can do the same counter work and can also approve customers, manage staff accounts, and monitor paid utang, activity logs, daily and weekly collections, the customers with the highest balances, and an audit log of who created, approved, updated, or deleted a record.

All records are stored in a MySQL/MariaDB database (run through XAMPP) that the Java application reaches through JDBC. Passwords and recovery PINs are never stored in plain text; both are salted and hashed before they are saved.

Objectives

- Keep the first name and last name in separate fields and validate empty names, contact numbers, usernames, passwords, and the recovery PIN.
- Require the customer to confirm the password and the recovery PIN during registration, with a Show option on each field.
- Hold every new customer registration as pending until the owner verifies a valid ID and approves it.
- Support partial and case-insensitive search of customers by name or account ID.
- Show the full text of a long date or note in a popup when the user clicks the cell.
- Let staff add an optional note to every utang or payment they record.
- Enforce role-based control in the service layer so that customers and staff cannot perform owner actions.
- Store every transaction and every important action in the database so the owner has a reliable audit trail.

Significance

The project shows encapsulation, inheritance through the Swing frames, and role-based access control in a system that a small store can actually run. It gives the store owner a record that is searchable, shared, and traceable, which a paper notebook cannot provide.

Part 2

User's Functionality

Utang Tracker has three users: Owner, Staff, and Customer. Each user signs in with a username and password. A customer who forgets the password resets it with the username and the 4-digit recovery PIN. Staff and owner accounts cannot use PIN recovery; they ask the owner to reset the password.

User	Functionality
Owner	Manages the store and has full access to every screen. • Adds utang and records payments when at the counter. • Reviews customer accounts with separate First Name and Last Name columns and searches by ID or name. • Approves pending customer registrations after checking the valid ID, updates customer details (name, username, contact number), and deletes customer accounts. • Resets a customer's recovery PIN. • Creates staff accounts, changes staff passwords, and activates or deactivates staff in Staff Management. • Reviews paid utang, activity logs, the daily report, weekly collections, the customers with the highest balances, and the audit log.
Staff	Works the counter and records utang and payments. • Searches a customer by ID or name. • Adds utang (item, quantity, unit price, and an optional note). • Records a payment (with an optional note). • Cannot open the owner reports, audit log, pending approvals, or staff management.
Customer	Registers an account and views their own balance. • Registers an account with first name, last name, contact number, username, password, and a 4-digit recovery PIN (password and PIN are each confirmed). • Receives a customer ID (CUST-#) and presents a valid ID to the owner for approval. • After approval, views only their own unpaid balance and transaction history. • A long date or note opens in a popup.

Account Rules

Field	Rule
First and last name	Required, at least 2 characters each, kept in separate fields.
Username	4–20 letters, numbers, dots, or underscores; must be unique.
Password	8–12 characters with at least one capital letter, one number, and one special character; must match its confirmation.
Contact number	10–13 digits for customers (staff may leave it blank).
Recovery PIN	Exactly 4 digits, entered twice at registration; stored hashed.
PIN lockout	Five wrong PIN attempts lock the PIN reset for that account for 15 minutes.

Part 3

System Process Flow Diagram

The diagrams below show the login and role-based flow, the customer registration and approval flow, and the forgot-password flow.

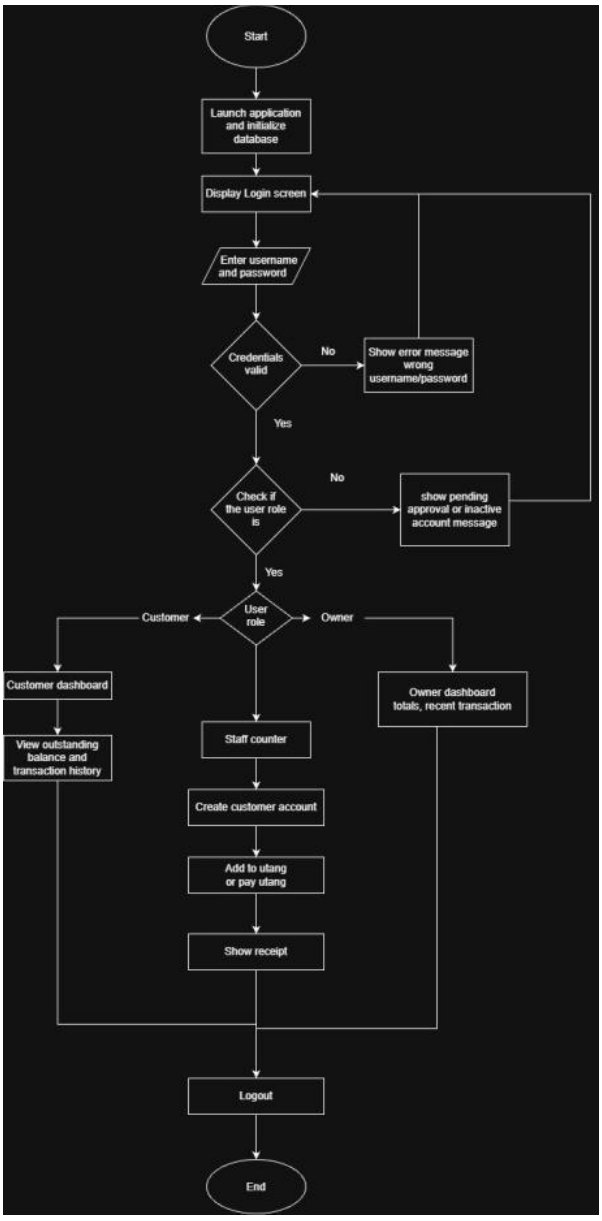


Figure 1. Login and role-based flow

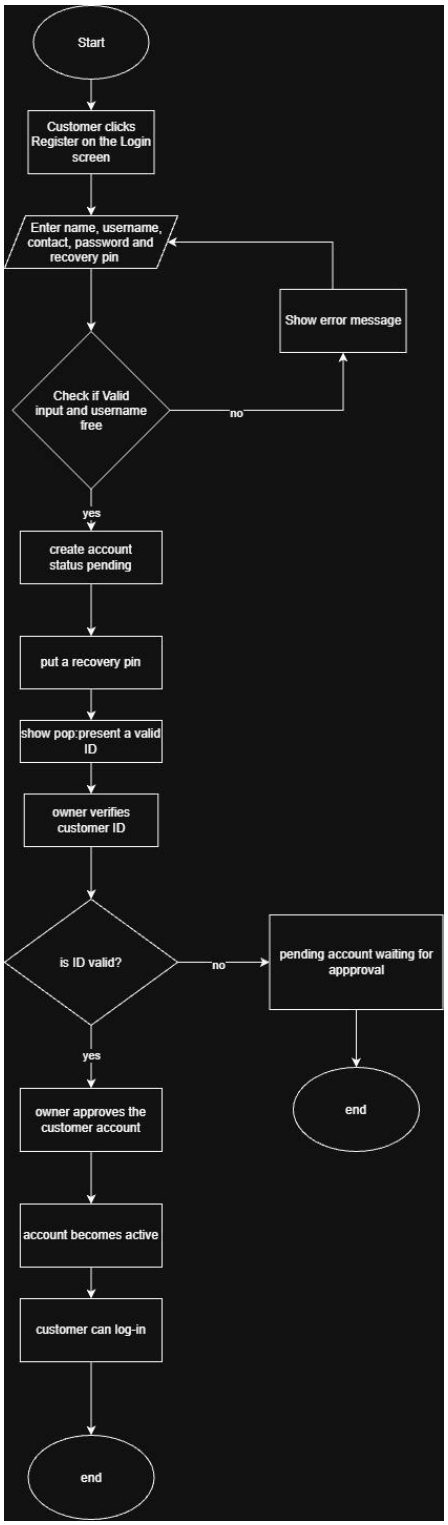


Figure 2. Customer registration and owner approval flow

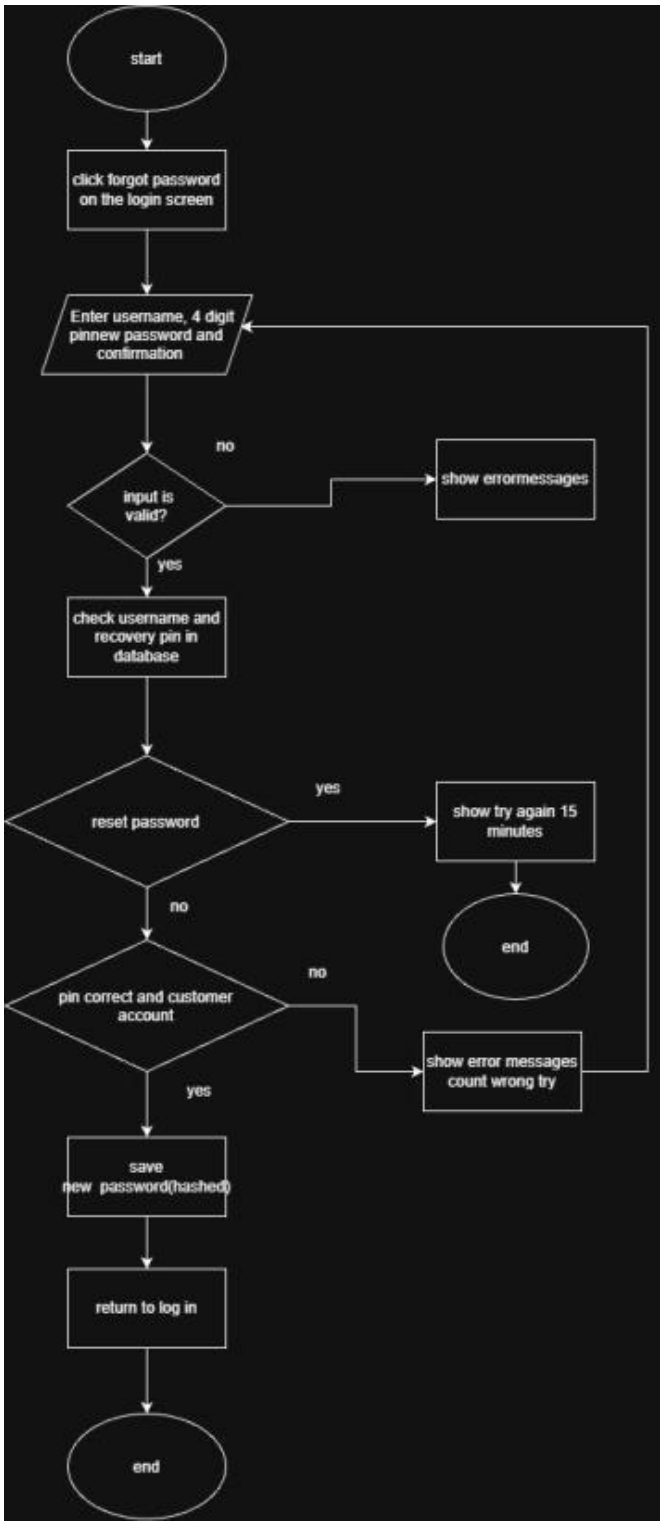


Figure 3. Forgot password (recovery PIN) flow

Part 4

Code Snippet

Utang Tracker System Package

Utang Tracker is organized into packages for its database, models, services, interface, resources, and utilities. The database package connects to MySQL/MariaDB through JDBC using the settings in db.properties.

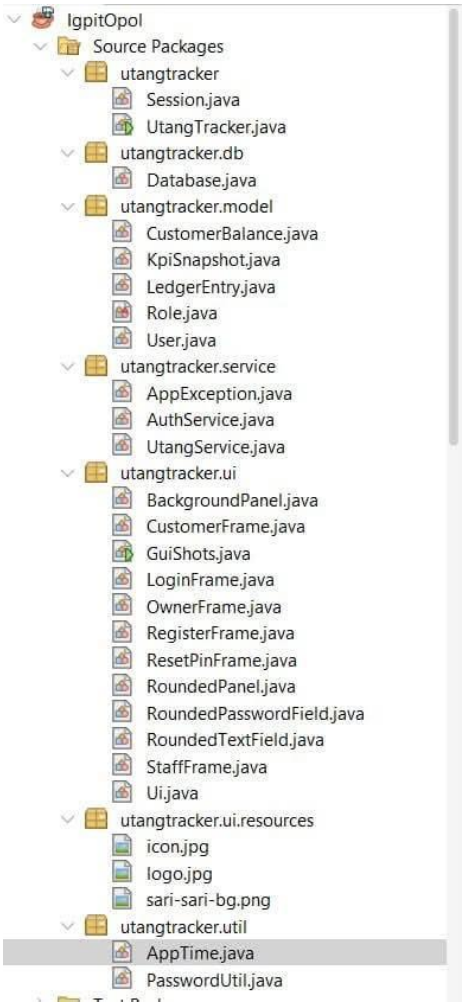


Figure 4. Project package structure

Database

The database igpitopol_utang_v2 contains the tables users, customer_accounts, transactions, activity_logs, paid_utang, pending_approvals, staff_management, reports, and audit_log. Triggers on the transactions and users tables keep the related tables up to date, the stored procedure sp_refresh_balance recomputes a customer's balance from the transactions that are not voided, and sp_generate_report saves the daily and weekly totals.

System Tests

The following snippets come from the SmokeTest class, which checks the main rules of the system without opening the interface.

Login and role test: This part is where a customer logs in and the check confirms the role as CUSTOMER.



```
// --- ledger basics ---
Session.set(auth.login("maria", "maria123"));
check(Session.get().getRole() == Role.CUSTOMER, "maria should be a customer");
```

Figure 5. Login and role test

Customer permission test: Customers are denied owner actions (approve a customer, create staff, view the pending list). Then staff log in and look up Maria's customer ID.



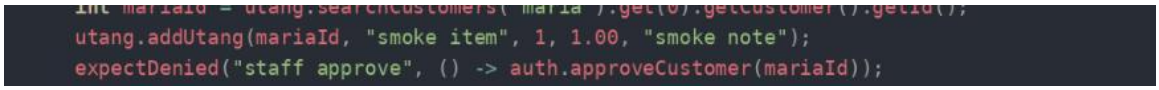
```
double before = utang.myBalance();
check(before >= 45, "Expected Maria balance around 45, got " + before);

// --- customers cannot do owner actions ---
expectDenied("customer approve", () -> auth.approveCustomer(1));
expectDenied("customer create staff", () -> auth.createStaff("Bad", "Actor", "bad.actor", "Passw0rd!",
expectDenied("customer pending list", () -> auth.pendingCustomers());

// --- staff can record utang but not owner actions ---
Session.set(auth.login("staff", "staff123"));
int mariaId = utang.searchCustomers("maria").get(0).getCustomer().getId();
```

Figure 6. Customer permission test

Staff permission test: Staff add a utang record for Maria (item, quantity 1, price 1.00, note). Staff trying to approve a customer is denied.



```
utang.addUtang(mariaId, "smoke item", 1, 1.00, "smoke note");
expectDenied("staff approve", () -> auth.approveCustomer(mariaId));
```

Figure 7. Staff permission test

Registration and approval test: The owner logs in and the total unpaid KPI must be positive. A newly registered customer is pending and cannot log in, and the error message must mention "approval".

```
// --- owner ---
Session.set(auth.login("owner", "owner123"));
check(utang.kpis().getTotalUnpaid() > 0, "Owner KPI unpaid should be positive");

// --- registration -> pending -> approval ---
Session.clear();
String user = "smoke" + (System.currentTimeMillis() % 100000);
auth.registerCustomer("Smoke", "Tester", user, "Passw0rd!", "Passw0rd!", "09171234567", "4321");
try {
    auth.login(user, "Passw0rd!");
    throw new IllegalStateException("pending customer should not log in");
} catch (AppException ex) {
    check(ex.getMessage().contains("approval"), "pending message should mention approval: " + ex.getMes
}
```

Figure 8. Registration and approval test

PIN lockout test: The 5th wrong recovery PIN must fail with a "15 minutes" lock message. The account stays denied even with the right PIN, owner and staff accounts cannot use PIN reset, and a new customer can reset a password with the correct PIN.

```
try {
    auth.resetPassword(user, "0000", "Newpass1!", "Newpass1!");
    throw new IllegalStateException("5th wrong PIN should fail");
} catch (AppException ex) {
    check(ex.getMessage().contains("15 minutes"), "5th wrong PIN should lock: " + ex.getMess
}
expectDenied("locked even with right PIN", () -> auth.resetPassword(user, "4321", "Newpass1!
expectDenied("owner PIN reset", () -> auth.resetPassword("owner", "1234", "Newpass1!", "Newp
expectDenied("staff PIN reset", () -> auth.resetPassword("staff", "1234", "Newpass1!", "Newp

// --- PIN reset success path on a fresh customer ---
String user2 = "smokeb" + (System.currentTimeMillis() % 100000);
auth.registerCustomer("Smoke", "Second", user2, "Passw0rd!", "Passw0rd!", "09171234567", "43
auth.resetPassword(user2, "4321", "Newpass1!", "Newpass1!");
```

Figure 9. PIN lockout test

Part 5

Graphical User Interface

Login: The entry point. Users sign in with a username and password (with a Show option). It also links to Forgot Password and Register.

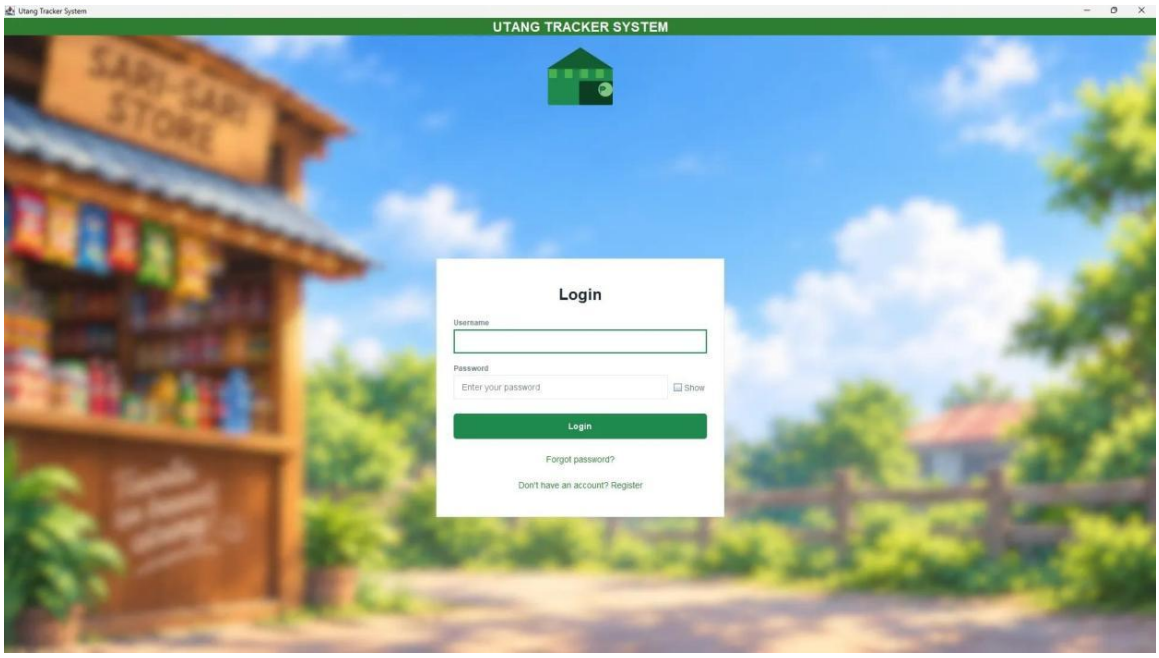


Figure 10. Login

Customer Registration: A sign-up form for new customers (first name, last name, contact number, username, password and confirm password, 4-digit recovery PIN and confirm PIN, each

with a Show option). The account stays pending until the owner verifies a valid ID and approves it.

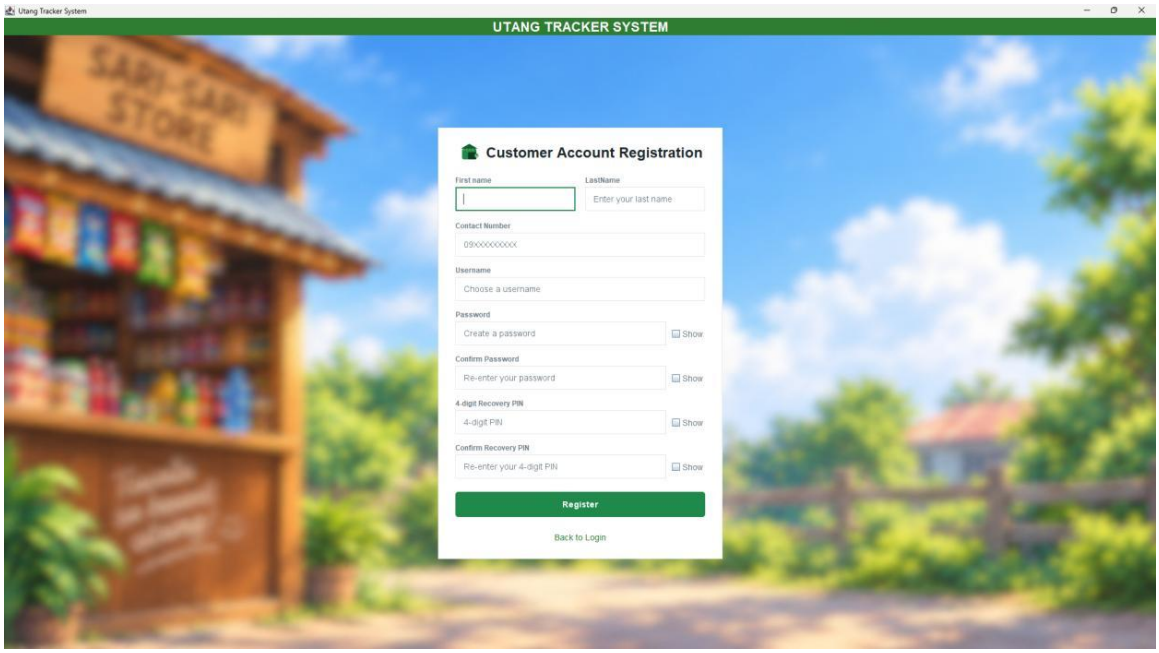


Figure 11. Customer Registration

Reset PIN (Forgot Password): Lets a customer recover the account with the username and recovery PIN, then set a new password. It locks for 15 minutes after five wrong PINs.

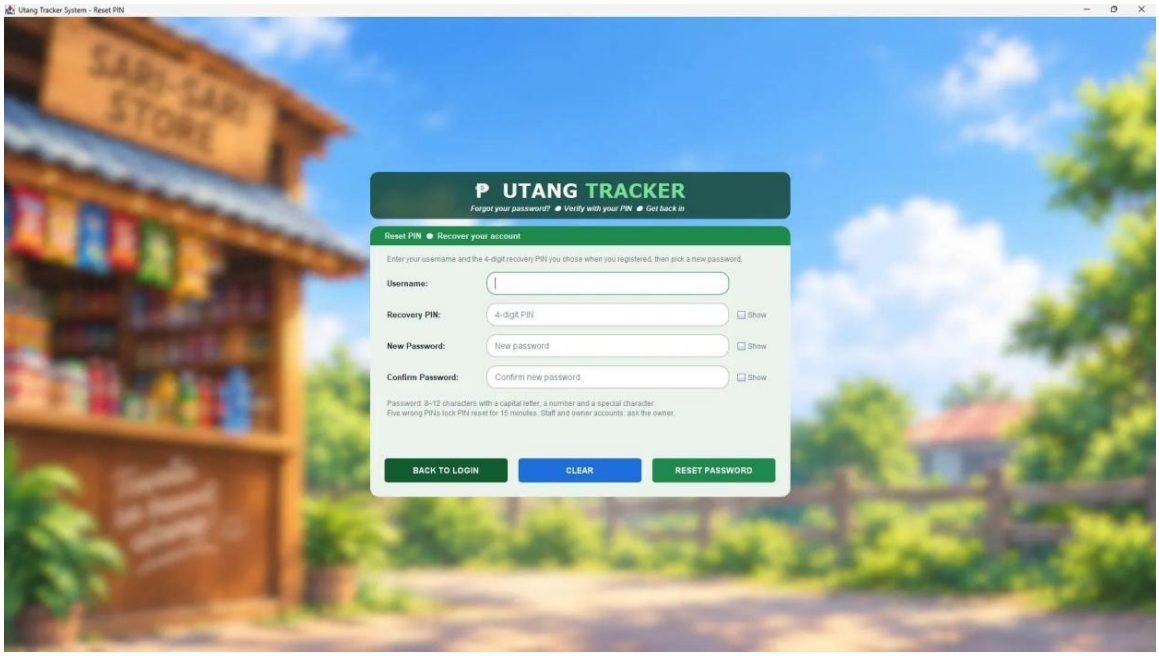


Figure 12. Reset PIN (Forgot Password)

Customer Portal: A customer's view of their total unpaid balance and transaction history (items borrowed, payments, notes, and paid/unpaid status).



Figure 13. Customer Portal

Staff POS: The staff's counter screen. Staff find a customer, then add utang or record a payment, with an optional note. Recent transactions are shown below.

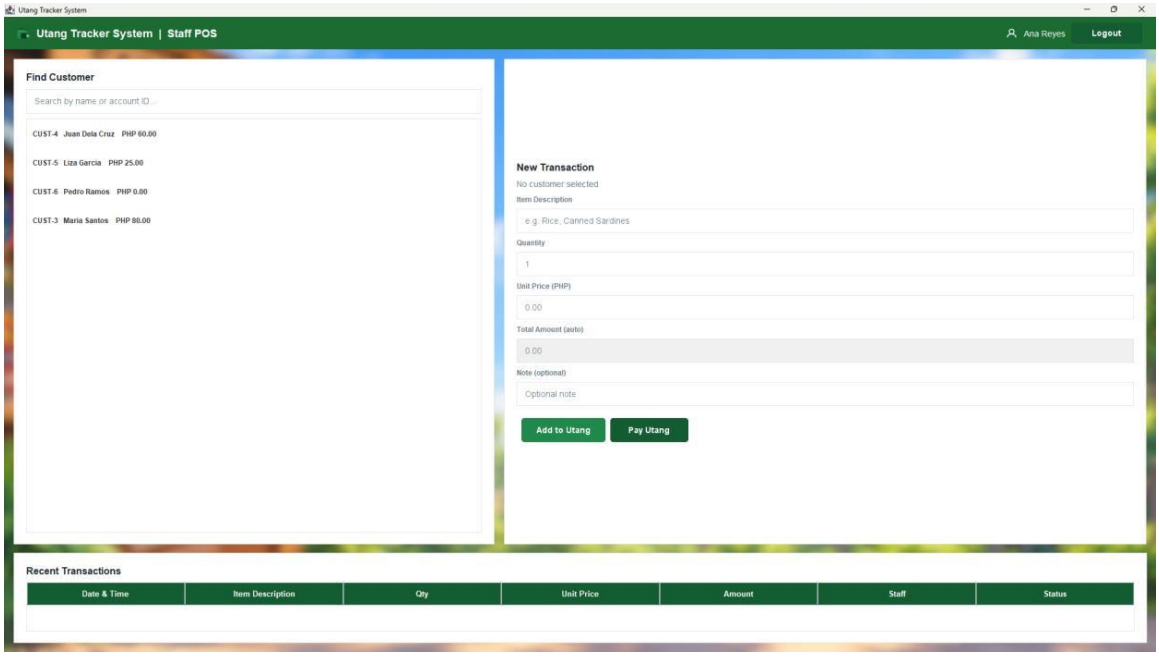


Figure 14. Staff POS

Owner Counter (Dashboard): The owner's main screen, showing total outstanding utang, active customers, and transactions today. It has the same add-utang and pay-utang tools as staff, plus Reset User PIN.

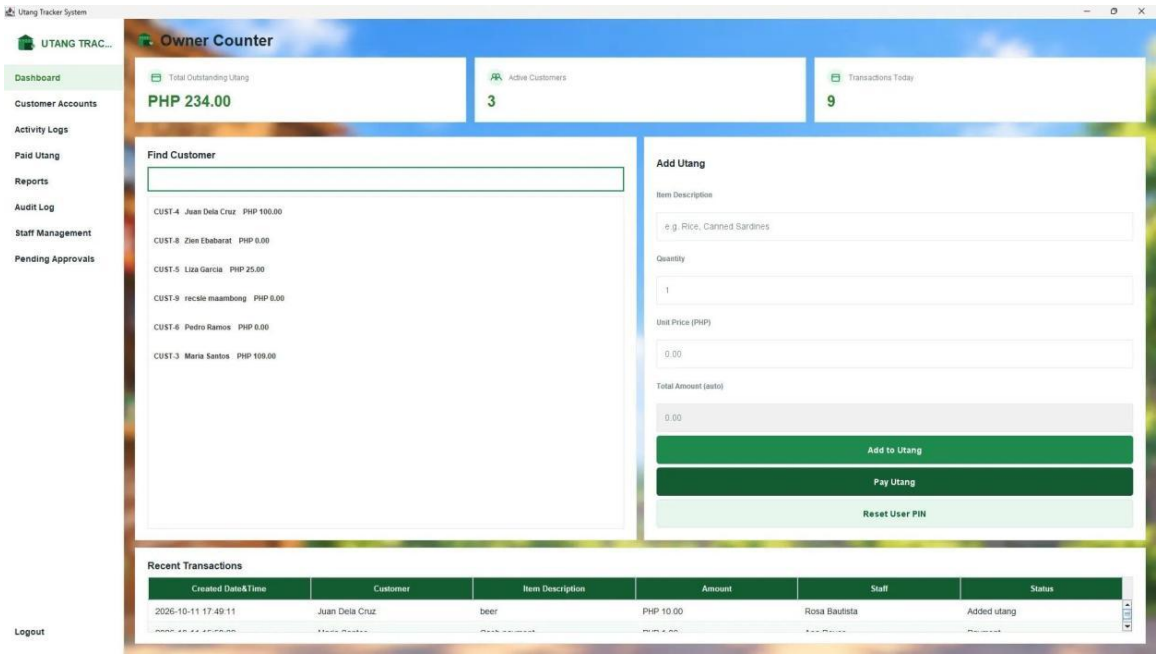


Figure 15. Owner Counter (Dashboard)

Customer Accounts (Owner Command Center): Lists active and pending customers with their balances. The owner can approve registrations and update or delete customers.

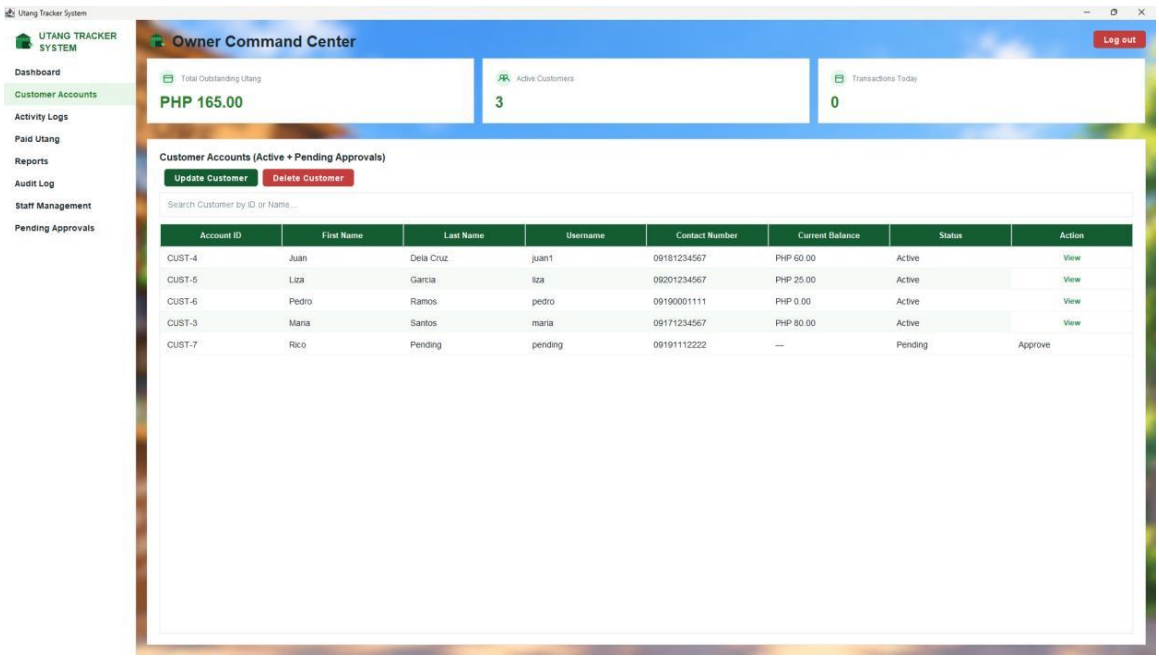


Figure 16. Customer Accounts (Owner Command Center)

Activity Logs: A detailed record of all utang and payment transactions (customer, staff, item, amount, note), filterable by date.

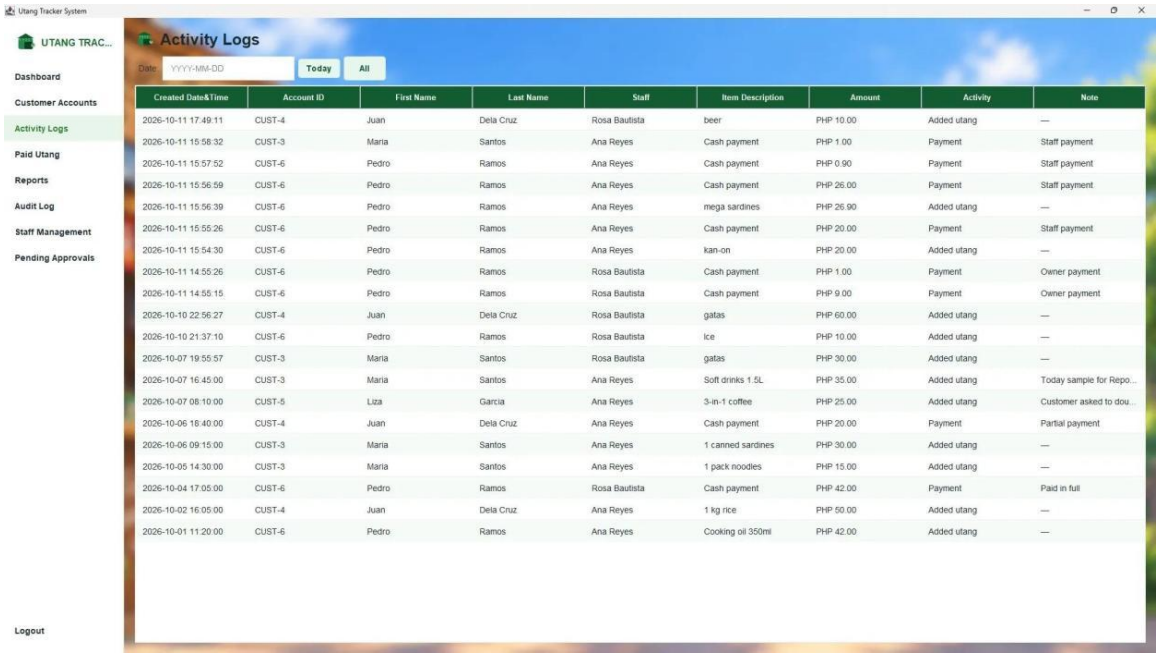


Figure 17. Activity Logs

Paid Utang: A history of payments received, showing who paid, who received it, the amount, and notes. It can be filtered by date.

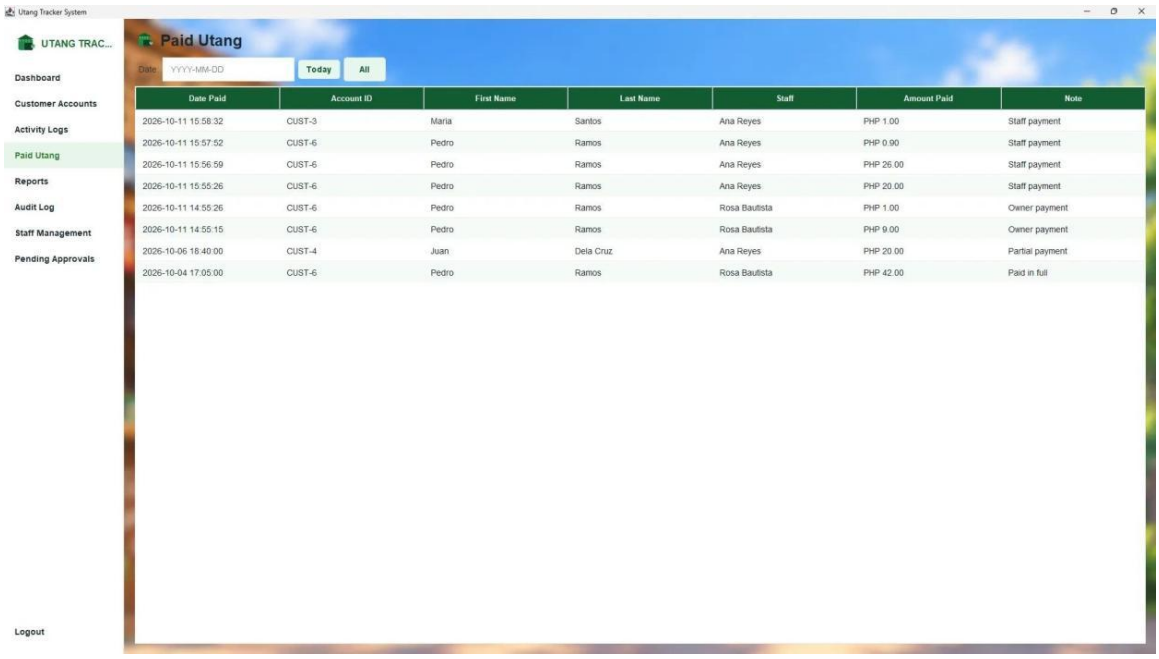


Figure 18. Paid Utang

Audit Log: A system-wide record of who did what and when (creating utang, payments, approvals, PIN resets, staff account changes), filterable by date.

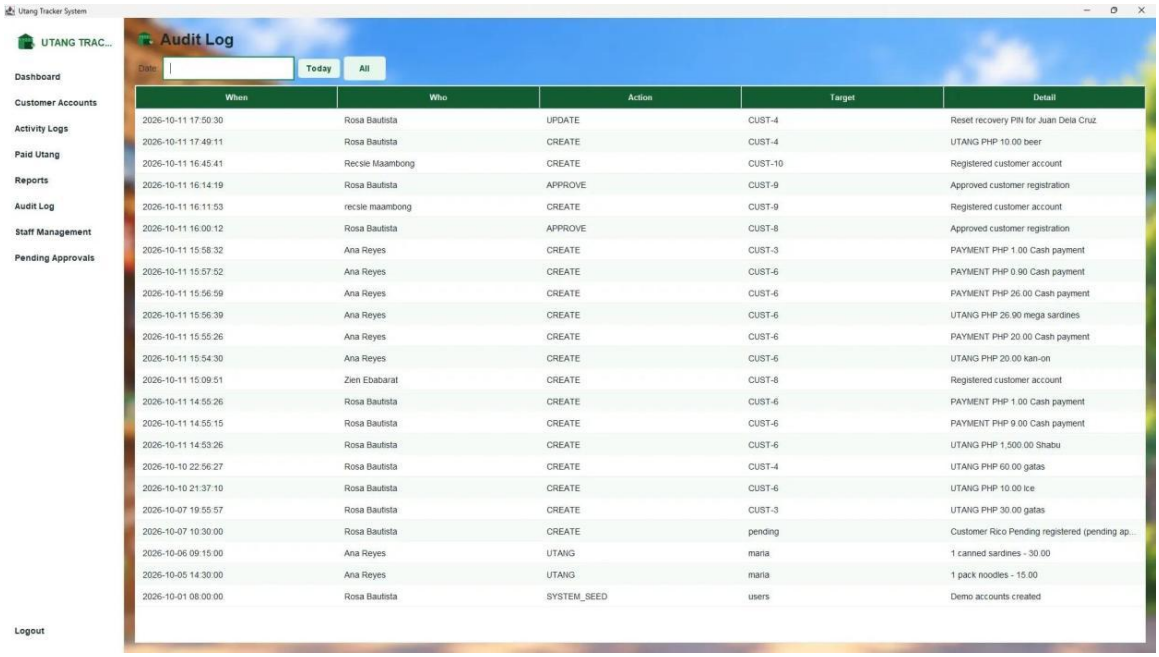


Figure 19. Audit Log

Staff Management: Shows each staff member's activity (utang entered and payments received) with a per-staff total. The owner creates staff accounts and changes staff passwords from this screen.

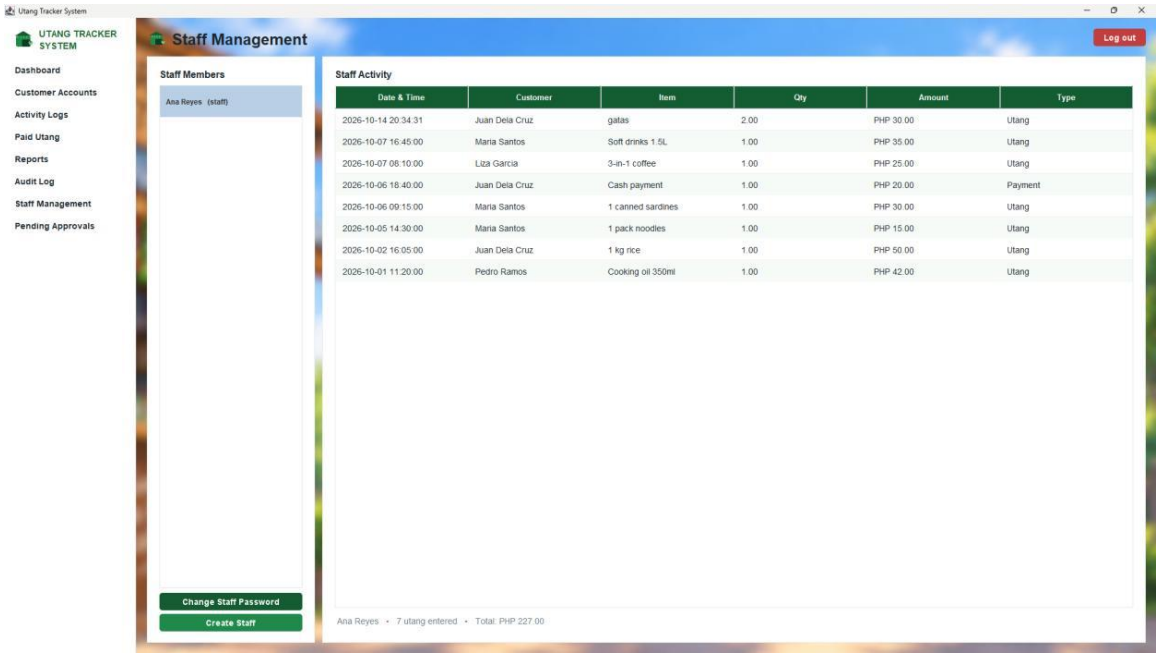


Figure 20. Staff Management

Part 6

Moments with Team



